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Risk Management planning of Metro Rail projects in India:

A Case Study of Bangalore Metro Rail Project

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Devendra Dutta Mishra is an IRSEE, basically an Electrical Engineer. He has worked in Indian Railways on many railway electrification projects involving construction, erection, testing and commissioning of electrical systems. Then he was on deputation on RITES Ltd. While in RITES, where he was Addl. General Manager and worked on the Delhi Metro Rail Project and also worked on the Bangalore Metro Rail Project as Dy. Chief Traction Power Engineer. Mr. Mishra is currently posted as Sr. DEN, S.W. Railways in Bangalore. He is also pursuing a PGPPM course at IIM Bangalore.

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Abstract

The paper aims to explore improvements in risk management planning for metro rail projects in India. The purpose is to recommend risk management measures that can be used for risk planning of metro rail projects in our country. The planning is envisaged to be carried out at all stages of the project life cycle.

The risk management measures are evolved out of the risk identification and analysis of major categories of risks like technical, construction, financial and operation and maintenance risks. Financial risks have been given emphasis in the paper, in relation to project financing. The risks that have emerged due to the challenges of shortage of talent, limited contractor capacity and rising labour and material costs have been given focus. The different types of risks faced by all project stakeholders, borrowers and lenders are considered.

The risk management measures that are a successful part of the Bangalore Metro Rail project are discussed, and these are also used for the recommendations. The authors have been a part of this project and their first hand experience and the project data has been put to use. Quantitative analysis has not been attempted as there is a hardly any quantitative data base for Indian metro rail projects. However, the risk mitigation measures brought out can form part of the policy framework of the Indian metro rail stakeholder organizations.

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1.0 Introduction: Risks associated with Metros

Metro Projects are a unique type of high risk infrastructure projects with long gestation periods. In this paper all the major risks associated with financing of Metro Railway projects have been studied/ investigated with a special emphasis on financial risks.

Infrastructure project financing has to encounter many types of major risks, viz. Technical, Construction, Political, Financial, Operation & Maintenance etc. These therefore are the broad categories under which the Risk identification, Risk Analysis and Planning of Risk Responses have been carried out in this paper. Only political risks have not being taken up, since the Metro projects are government owned and political risks usually take their own course.

There is a paucity of documented quantitative data for metro projects and hence the authors have not carried out any quantitative analysis in the paper. So for this paper, critical Risk Analysis based on the first hand experience of the authors has been carried out for the identified risks. A risk engineering approach need not necessarily be about measuring risks, and it need not be probabilistic. It should be concerned with what might happen and what should happen. ^{1Chap 1}

The Bangalore Metro Railway experience has been studied in dealing with the identified risks. For each of the broad category of risks, risk response planning has been carried out separately. The objective of this study is to recommend Risk management measures which can be applied at all stages of an Indian Metro Railway Project life cycle.

2.0 Risk Identification and Analysis

The need for risk analysis of potential projects and investments is particularly apparent when projects involve large capital outlays, unbalanced cash flows, requiring a large proportion of total investment before any returns are obtained. ¹

2.1 Technical Risks:

2.1.1 General

Metro rail projects are executed using sophisticated and highly complex technology, associated with high lead time and low technical knowhow in procurement of the technology and also in the equipment and the commissioning and operation of the system. These risks and uncertainties have to be accepted by the project planners and the managers. Notwithstanding these risks, they also provide for opportunities of the competitive advantages over others due to ownership of improved and innovated technology in the construction, operation and maintenance of the project.

2.1.2 Technical risk identification and analysis

Technical risks for Metro Railway projects are effectiveness of proven/ new technologies, tender management risks, design related risks, schedule and programming related risks, Estimation risks, risks affecting Scope, industrial risks etc.

Technical risk is also minimised before lending takes place by obtaining experts reports on the proposed technology. Technical risks are managed during the loan period by requiring a maintenance retention account to be maintained to receive a proportion of cash-flows to cover future maintenance expenditure.

The choice between established and proven technologies and new emerging technologies is a difficult one. Although new technologies may be risky, engineers tend to use them on new projects.

Different delivery methods like Design-Build (DB) or Design-Construct (DC), Design-Build-Operate-Maintain (DBOM), Build-Operate-Transfer (BOT) , have different type of risks inherent in them,

When the various service providers are segmented, the owner has the most amount of control, but this control is costly and doesn't give each provider an incentive to optimize his contribution for the next service. When there is tight integration amongst providers, each step of the delivery is undertaken with future activities in mind, resulting in cost savings, but limiting the owner's influence throughout the project.

Cost Estimation Risks - Risk analysis of a project cost estimate has a number of benefits. It provides an indication of the reliability of the cost estimate and the adequacy of the contingency allowance^{1 pp170}. The cost estimation at the tender stage carries the risk of over or under estimation

Technical risks also arise from the preferred design options for the project, whether multiple designs or a standardization of the design across the project.

Schedule and Programme related Risks

The schedules may under or over estimate time or cost applicable for the tender stage or the construction or operation stage activities. This will then result in incorrect determination of the Project completion time, which has the potential of delaying the project.

The identification/breakup of detailed activities and their construction planning is most important, to complete the construction work, without any time or cost overrun. An incorrect detailing of the project through a under developed Work Breakdown structure will result in some

of the project elements being left out at the planning stage. This will also affect the project work adversely.

Industrial risk taken by the Contractor

To keep the project on schedule, the contractors may resort to taking of some industrial risks, like ⁵entering into the production phase before the closure of the design phase. Even the prototype manufacturing and testing may be dispensed with, taking advantage of the learning curve of similar previous projects. While this may be acceptable in principle and inevitable from practical considerations, the risks associated need to be weighed by the Owners/ Consultants before allowing the contractor to take these risks. Many contractual issues may also be involved.

Risks affecting Scope

There may be scope creeps in the project due to changes in design or technology, alignment, political reasons etc.

2.1.3 Technical Risk Management for Bangalore Metro rail Project

In case of Bangalore Metro Rail project, the Detailed Project Report, which was prepared by Delhi Metro Rail Corporation and RITES, the well-known Consultants in planning and construction of Metro rail system in India was accepted by government in year 2003, recommending proven and latest technology, similar to that used in most modern metro rail systems in the world, such as Hong Kong, Singapore and Bangkok Metro rail systems. Before start of the construction of the project in year 2006, the technology recommended in DPR was further reviewed and critically analysed by Technical Experts of General Consultants(a consortium of RITES-OC-PBI-SYSTRA, appointed as General Consultants (GC) for the project in Aug 2006), who have global expertise of Metro Rail Systems. Thus the technical risks in the project were minimized.

Traffic diversions, utility diversions, geological investigations were carried out in the tender award period by the General Consultants. Most of these works were sub-contracted so that sub-contractor expertise could be utilized and comprehensive reports were taken from them. Detailed

design consultants (DDCs) were also appointed for preparation of the designs, Bill of Quantities, and for getting approvals from the statutory authorities. The DDCs are also continuing, for construction support, and this transfers effectively some of the design risks to these entities. Different traffic projection scenarios were studied by the General Consultants.

A Schedule was prepared by the BMRCL at the initial stage itself which proved to be very useful for the planning of the project. It had detailed time estimates for the tender process for all the major works envisaged at that time, and the durations for constructions were taken as a lump sum, based on other Indian metro projects. During construction, the provision of sufficient floats in detailed schedules was ensured, to minimize the schedule risks. Other time extension risk minimizing used through the tool of schedule were a detailed work breakdown, integration of all schedules through an EPS (or Enterprise Project Structure) based programming software called Primavera v 6.2. Recovery schedules and catch up programmes are being regularly prepared.

2.2 Construction Risks

2.2.1 General

If planning for construction risks is done in time, then all alternative decisions and cost associated with them can be known before hand, it can help during execution process to take appropriate alternative decision and mitigate risk.

Delivery and Completion Risk identification and their allocation is a vital part of the risk management of any project. This phase carries the greatest risk for the financier as well. Construction carries the risks and uncertainties that the project will not be completed on time, on budget, or jeopardized at all, because of various construction difficulties and challenges. Such delays or cost overruns may delay revenue operations, and thus default in loan repayments and cause interest and debt to accumulate. This may also jeopardize contracts/agreement for the sale of the project's output and services to the government/customers/public.

2.2.2 Construction risk identification for Metro railway projects

Some of the major construction risks on a Metro Railway project are presented below:

Land Acquisition

A major risk for Metro rail project is the one associated with land acquisition. This would impact project scope, construction time schedules, cost and could even result in part or full abandonment of the Project. Sometimes litigation and court cases are taken recourse to by the affected parties, for settlement of land acquisition disputes.

Environmental Issues: The most critical environmental issues are those related with cutting of trees infringing the metro alignment, noise pollution , risk of damages and vibration risks to adjoining buildings, especially the old historically important heritage structures, and affecting aesthetics of the city due to the metro rail structure.

Mobilization of Plant & Machinery & Equipment: The Plants and equipments such as Tunnel Bore Machines- for the construction of tunnels , Rolling Stock (metro coaches), Power Supply Plants and Equipment and Third Rail System, Signaling System, Automatic Train Control and Operation system, Automatic Fare Collection System etc are not available indigenously and they have to be procured from abroad-mostly from the European countries as well as Japan and Korea. The mobilization of these plants and equipment is associated with various procurement issues and the long lead time. In addition, mobilization of the manpower and experts with technical expertise and skill to handle the operation, erection of these plant and equipment is also to be done. These issues add significantly to construction risk and uncertainties of the project.

Safety During Construction: The construction of heavy elevated civil structures- launching of overhead segments and girders of the elevated structure, along the median of the

roads, in the very middle of densely populated city, with high uninterrupted flowing traffic causes various issues related to the safety of the general public, as well as of the working staff associated with these construction activities. Thus, a very safe and secured construction methodology is essential during the construction of elevated structures and the tunnel construction works. The movement of road trailers, loaded with the structure segments, the launching of these segments to the overhead structure, the construction of overhead slabs of station buildings over the roads, the tunnel cutting/boring are some of the most critical activities during civil construction works.

Traffic Management/ Diversions: Metro Rail alignment requires deliberating a lot of traffic management schemes and the diversion of traffic during the civil construction stage. It is one of the greatest challenges in the metro rail project construction. In this regard, lot of coordination with traffic authorities, local bodies, and public is essential. There may be delays in decisions regarding the schemes due to difference in opinion on the layouts and responsibility of implementation.

Quality Control

Quality cannot be compromised on a metro rail project. The challenges lie in achieving quality in the face of shortages of time and cost. Delays in construction approvals, delays in tests, risk of bad workmanship, risk affecting structural and other failures, third party proof checks etc. all add to the risk of time delays and cost overruns.

Force majeure risk

This is the risk of events which render the construction or operation of the project impossible, either temporarily (e.g. minor floods) or even permanently (e.g. complete destruction by fire).

Mechanisms for minimizing Force Majeure risks include: (a) conducting due diligence as to the possibility of reducing the relevant risks; (b) allocating such risks to other parties as far as

possible (e.g. to the builder under the construction contract); and (c) requiring adequate insurances which note the financiers' interests to be put in place.

2.2.3 Construction Risk management for Bangalore Metro Rail project:

Land acquisition was meticulously carried out in areas like CMH Road through negotiations and legal measures. Changes in route alignments were resorted to wherever necessary, to avoid land acquisition.

Tree cutting became a major issue in Jayanagar area, disrupting construction work for some months. But finally this issue was resolved successfully and construction work re-started.

The P&M requirements for various contracted works have been well documented in the tender documents, including the numbers required, specifications, suggested brands etc. This has been used for the technical evaluation of the tenders as well. After the contract award, the critical P&M proposed by the contractor is reviewed. During the progress of the project, if the works are in delay, then the deployment of additional requirements of P& M is planned to recover or catch up the works to the extent possible.

Some of the typical construction risks(in addition to the usual risks common to infrastructure projects) that were identified by the Management are Complexity due to Contracts, Complexity due to Phasing, Urban Environment, Multiple dependencies for System re installation & delays, Discontinuous Viaduct , Time frame expected and Political Aspirations, & Project Scope Change/creep, Organizational relationship & Effective Communication, Decision Making.

Some functional areas have been identified for special risk assessment like Tunneling works, Segment launching, third rail procurement, cable laying on viaducts etc.

The launching of girders is being carried out during night time. Traffic marshals are being adequately deployed on the project sites to control safety.

Traffic diversions are being planned site by site, and in advance of the actual construction works to be carried out. At many locations, temporary roads were made to divert the traffic till such time that construction was completed to a certain level and then normal traffic was restored.

The tunneling work is very hazardous and tedious work so construction risk is very high in the project. The associated risks to the tunnel works are identified as following:

Geological risks for tunnel in subsoil with high water table, risks associated with safety at work and disturbance to surface traffic in municipal areas, stability of submerged tunnel, tunnel ventilation problem, too much of noise pollution, traffic accidents and fire breakouts for tunnels during operation..

The phased opening of the project has been envisaged, which is the opening of the eastern leg (R1) first, followed by the northern leg (R3) thereafter Underground (UG) and then immediately the Western and Southern legs.

Due to various reasons beyond control, the project was delayed. To alleviate the risk of delayed opening of the reaches the Client Consultant team is planning to change the phasing plan, to give the people of all areas of Bangalore early access to Metro services. This is supported by the fast progress of works on say the southern route, when compared to other routes.

Interface management plans and inclusion of interface activities in the schedules are being used on the project to ensure smooth flow of works once more than one contractor start working in the same area at the same time. At the peak of construction works, about 4 to 5 contractors will have to be working together in the same workspace in a coordinated manner. All contractors working on the project have dedicated interface managers who plan the interface issues, both at design and construction stage. This work is also coordinated by GC/BMRCL.

Quality checks are being carried out at the suppliers' factories before allowing the shipping of critical components. The source or the suppliers for the construction components are already specified at the tender stage itself.

2.4 Financial Risk Management

2.4.1 Financial risk identification and analysis

Financial risks have been studied under the heads of project cost, currency related risks, commercial risks, project financing loan structures etc. The financial risks borne by lenders and borrowers have also been dealt with separately.

Currency risks include the risks that: (a) a depreciation in loan currencies may increase the costs of construction where significant construction items are sourced offshore; or (b) a depreciation in the revenue currencies may cause a cash-flow problem in the operating phase.

There is a risk that the final project cost may escalate due to rise in prices of resources, Change in price indices/ tax structures, multiple modifications of design or construction methodologies. There is a risk of requiring suppliers to provide support after completion at potentially less than beneficial cost etc. At different stages of the Project, assessment of revised completion cost, for the purpose of planning its funding pattern, drawdown schedule, financial closure of the project are needed, which has inherent risks to the borrower.

The loans for metro projects are mostly non- recourse loans, which are secured by the project assets and paid entirely from project cash flow, rather than from the general assets or creditworthiness of the project sponsors. Project lenders are given a lien on all of these assets, and are able to assume control of a project if the project company has difficulties complying with the loan terms.

Lenders do require that any risks to project revenues be allocated to, and mitigated by, the Project Company and/or sponsors.

To ensure project revenues sufficient to service project debt, lenders should include in the provisions of the project debt documentation a requirement that the sponsors inject the agreed funds into the project company to stabilize cash flow and reserves or top-up the shortfall in pre completion project revenues.

Since ODA programs are implemented in developing countries, credit risk is considerable, and for metro projects, interest risks are more because loans are extended for a long term. However, the government will very likely take necessary financial measures including additional investment or transfer of grants if the risks become evident, so that there will be no effects on operations of JICA as an execution agency of ODA programs. Its equity capital is solid with strong risk resilience. Furthermore, as regards the loan assistance account, JICA has created an adequate risk management structure based on characteristics of its portfolio. It quantifies various risks as a part of integrated risk management, and verifies capital sufficiency as well as impacts on financial performance based on stress tests and scenario analysis

2.4.2 Analysis of financial risks

Risks inherent in financing models for Metro rail Projects in India

Two project financing options are used for Metro Rail projects – EPC and PPP. The usage of these models in the currently ongoing Metro projects is presented below:

Bangalore and Chennai are two metros that are progressing well on the EPC route. Kochi metro rail, which was earlier planned as a PPP project, has now rolled back to EPC route.

Recently two major projects—Mumbai Metro Line 2 and Gurgaon Metro went to private sector. For Mumbai Metro Project, the then-prevailing economic slowdown had made many a private sector entrepreneur wary of investing and sharing the risk of such a large and unprecedented project. Finally, Reliance Infrastructure, the only bidder put in its financial bid and was awarded the project.

For the 22-km airport expressway link, Delhi Metro Rail Corporation will do the entire construction (the cost-intensive portion of the project) while private entrepreneurship (a consortium of Reliance Infrastructure in this particular case) will be responsible for subsequent aspects like tracks, rolling stock, signaling, and eventual operations and maintenance.

Reportedly, Hyderabad Metro is planned on the PPP model.

So it is seen that there both models are in use on Indian Metro projects, but it is predominantly EPC which is being used, while PPP model is slowly gaining popularity.

At least 20 tier-II cities propose to have metro rail systems, but it is not yet clear if these cities could generate the minimum passenger traffic to make metro rail viable. And even if they are viable in some cases, it is not clear if the private sector would be sufficiently enthused in developing them, as they would be in, say, metropolitan and tier-I cities.

If India expects funding from multilateral funding agencies like JBIC, JICA and World Bank for metro rail projects, the EPC route would be a default choice as such agencies usually do not finance PPP projects. Hence the choice between EPC and PPP routes is a difficult one.

2.4.3 Financial Risk management in practice for Bangalore Metro Rail Project

The costing of the Bangalore Metro project has been done in a detailed manner and cost reduction exercises are being carried out. Recently, for the underground tender, the tenders were divided into many parts, so that the overall cost is reduced. Cost control measures are in use.

Liquidated damages, performance bonds, retainage accounts, warranties and contingency funds are all being successfully used on the different contracts of the project.

Prior to contract award, the tender cost estimates were used for cost management and later, when all major tenders were awarded, realistic completion costs have been arrived at.

The revised completion cost as on date is more than Rs. 11000 crores. DPR cost has increased due to the following major reasons:

Due to the increase in scope of work, additional features not envisaged in GOI sanction, ⁶ Specification variations and design modifications, due to price variations.

Some of the specific reasons for cost increase are – Alignment and Formation for underground section, Cost of land acquisition, change in cost of station buildings, Lifts and Escalators etc.

In the drawdown, some significant amount (say 500 crore) has been earmarked for withdrawal during 2013-14, since some of the bills could get settled that late, though physical work might have been completed a little earlier.

Limited recourse loans are under disbursement on the project. Since the project cost has escalated, new sources of financing are being explored from lenders like ADB, AFD, KEXIM (Import Export bank of Korea), these are limited/ non recourse loans and hence risk management measures for these new loans are being worked out.

Although the major part of the project is being successfully executed on the EPC model, one of the works, viz. Swastik elevated station has recently been awarded on PPP basis to Mantri Developers, since the opening of R3 section will need Swastik station to be completed on time,

Risk identification for Bangalore Metro based on Project Stakeholders:

Different type of risks are allocated to different entities, hence JICA., BMRCL, GOK and GOI all carry different type of risks. The Project Consultants GC also is affected by certain risks.

These organizations have all developed Risk mitigation measures to deal with their respective risks. Abandonment of the project is a common risk for all parties. A study of the risks and the risk mitigation methodologies and strategies in use is described in the following paragraphs:

Risks for JICA

Borrower using proceeds of the loan for works other than the project. Documents evidencing usage needed to mitigate this risk

Cash flow not matching Amortization schedule⁴ Schedule 3,

Risks implicit in the tender review like for prequalification, technical evaluation etc.

Since the number of JICA awarded contracts is few, risk of non- JICA contracts interfaces affecting the progress of JICA awarded contracts.

Increase in Category D components i.e. “Contingencies” which may carry a lower rate of interest.

Risks for GOK/GOI

To bear the entire cost of land through interest free subordinate debt from some other source than JICA, they will ensure that electric power is made available to the project on a no profit no loss basis.

Any cost escalation due to statutory levies and duties, exchange rate variation and price escalation will be borne equally between the project promoters⁵. Any change in the scope or delay beyond the approved project time cycle will be borne by GOI⁵.

Financing of cash losses and capital expenditure during operational phase and its requirements

To ensure price based measures to promote and facilitate metro ridership, as part of an integrated traffic integration plan for the Bangalore city with a view to ensure that the project ridership is realized.

Risks for BMRCL/GC

Funding arrangements if the cost of the project escalates beyond the initial funding arrangements

Catering for losses to the Borrower and/or the Supplier owing to any difference between the exchange rate to be used for conversion from the Japanese Yen to the currency used for actual payments to the Supplier (s) to the Japanese Yen at the time of disbursement.

Any change in the Japanese Government’s annual budgetary appropriations for JICA

Any changes by JICA in the “Guidelines for procurement under JBIC ODA loans or any other guidelines directly or indirectly affecting the Project.

Errors in distribution of Project cost i.e. categorization for allocation of loan proceeds

Errors in initial Project plans – like in DPR, Errors in estimation of Project Cost, in time schedules, and in Scope.

2.5 Operation and Maintenance Risks

2.5.1 Risk identification and analysis

These are risks that may affect the cash-flow of the project by increasing the operating costs or affecting the project's capacity to continue to generate the quantity and quality of the planned output over the life of the project. Operating risks include, for example, the level of experience and resources of the operator, inefficiencies in operations or shortages in the supply of skilled labour. In case of private operators, the usual way for minimising operating risks before lending takes place is to require the project to be operated by a reputable and financially sound operator whose performance is secured by performance bonds.

Traffic projections carry an inherent risk for Metro projects. Since the projected traffic figures may not get translated into actual traffic figures during the O&M phase, this will have a negative impact on the revenue generation and subsequently on the sustainability of the project.

One of the foremost causes of the traffic projection errors is the absence of an integrated traffic management plan, or there exist many missing links in the traffic management plans, a result of faulty coordination between government bodies. Commuters may not find Metro as an attractive option as there is no point to point service and no system of feeder buses to the metro stations.

Operational resources risk - These are the risks associated with sufficient resources availability with the clients to manage the operations of the project and to efficiently resolve any problems which may arise.

2.5.2 Operation & Management Risk management for Bangalore Metro Project

Operational expenditure has been projected realistically by BMRCL/GC, under major heads like manpower cost, energy cost, maintenance cost, insurance charges and administrative expenses. North and south extensions would need additional power consumption due to additional stations and additional number of rakes and this aspect has been considered in the planning.

Training needs are already being sorted out by BMRCL. Experienced staff recruitment process is already underway much before the opening of first reach of BMRCL

Also the actual power and the cost of the power may vary from the assumed figures depending upon the actual configuration of the power equipments to be installed in the Metro system, and re planning will be carried out as soon as the operations begin.

3.0 - Recommended Risk management planning measures for Indian metro railway projects and Concluding Remarks

Each Metro rail project gives rise to its own unique risks and hence poses its own unique challenges. In every case, the parties - and those advising them - need to act creatively to meet those challenges and to effectively and efficiently minimize the risks embodied in the project in order to ensure that the project financing will be a success.

After the risk identification and critical analysis that has been carried out earlier, and based on the risk management measures that are in use on the Bangalore Metro project, some risk management measures are being presented for general risk management on metro rail projects in India.

These are being presented in the same format in which the identification and analysis was carried out, viz. technical, construction, financial and operation and maintenance risks:

3.1 Technical Risk management measures

1. The planning of the project has to be started off as soon as the approval of the project comes through or even from the time of anticipated approval. The long gestation periods of a metro railway project can be utilized for planning all aspects effectively so that implementation can commence very soon after award. The measures can include marketing initiatives with the lending agencies, weighing of the project financing options, decisions on the number of work packages, the forms of contracts for these packages, planning for the tender management process, planning the organizational set up, determining the availability of the proposed experts etc.
2. Design and build contracts should be preferred. The issue of Professional Indemnity (PI) insurance must be addressed as D&B Contractors may not carry the PI and in the event of a claim against PI there may be administrative difficulties in recovering any loss against insurance.
3. Following contract award, the P&M requirements have to be planned by the team of Clients, consultants and contractors and procurement processes initiated. Client, consultant and contractors should have consensus on the backup provision of costly capital equipment. the minimum equipment requirement has to be phased out optimally and accordingly the planning in terms of placing of purchase order, arrangements of transfer of the equipment from other sites, fitness and calibration certificates, availability of land in the works area to house the equipments etc, need to be planned.
4. The initial works like traffic and utility diversions, regulatory clearances, geotechnical investigations, environmental issues, working methodologies, the use of new technologies etc. carry inherent risks in terms of time and money. Hence, liaison with city authorities has to be put in place upfront in the project. There may be a need to change the alignment routes, which needs to be identified. A slight change in alignment has a cascading effect and leads to many technical changes in the project scope. Along with proven technologies, new technologies need to be embraced.

5. The scheduling exercise for the project must start just after the award of the project. The client can prepare summary level schedules, with whatever clarity is available at the point of time of preparation. Use of planning data from similar previous projects, industry standards, which are moderated to suit the project can be used. This will give a client valuable inputs to the time and cost requirement of the project. If there is a general consultant involved, he can take over the estimates at his mobilization. The tendency to leave scheduling exercise to the contractors (when they are mobilized) with the argument that the contractor is the one who will actually do the work and any schedules prepared earlier will be “way off the mark” is not correct and should be avoided. The preparation of schedules upfront also helps in preparing tenders and reducing scope creeps and in the review of the contractor schedules when they are to be contractually submitted.

3.2 Construction Risk management measures

1. During construction of Metro railways, the phasing of the project may need to be reviewed. To provide the customers earliest possible access to the public, the routes that have progressed fast need to be planned to open first. Accordingly, the work on these routes needs to be prioritized in comparison to other routes, which may have been the priority as part of the initial plan. This will affect major changes in train depot works, re allocation or redistribution of resources etc., but for the benefits of early revenue operation of the project, this may be a small price to pay. There may be instances when to do the same work along a particular stretch, there can be more than one sequence of completing the work. In such cases, it is also recommended that the locations where there are more risks, in terms of difficult geological terrain, appearance of uncharted utilities etc. should be taken up first. This will give benefits of using a learning curve approach, and also provide more time to tackle the less risky stretches, once the risk ones are done with and out of the way.
2. Interface issues are potentially very risky, since they need the consensus of many engineers, with their own preferences and compulsions. Interface planning is most important for a metro rail project. There would be interface issues at the design stage and

also at the construction stage. There has to be an insistence on the early deployment of an interface manager by all contractors, and the interfacing efforts need integration at the client/ contractor level. Differences in the interface documents like combined service drawings (CSDs) need to be resolved and signed off by all concerned parties, to avoid conflicts at the time of actual; installation and testing.

3. There are some functional areas which are high risk areas, e.g. tunneling works, third rail works, cable laying etc. Special attention needs to be given to these areas. Noise vibration studies, building condition surveys etc. performed by third parties are also needed.
4. Strict quality management at the construction stage is needed to reduce the operation and maintenance needs of the metro system. The testing and commissioning phase needs to be well planned out and strictly enforced, like the factory acceptance tests (FATs), type tests and integrated testing and commissioning. Systems like Automatic Train Control (ATC), Automatic Train Protection (ATP) should not be compromised. Training of metro staff will offset some of the manual risks involved in the metro operation. Operation and training manual planning and preparation have to begin at least six months before the train operations are to begin. Availability of sufficient number of crucial resources to manage the efficient operation of the project has to be ensured. If necessary, operators with experience on similar projects can be recruited in the beginning who can then train the local people for say a one year period. Contracts for the crucial train depot equipments will have to be awarded early because these have to be designed to the rolling stock (train) specifications. If this is not possible, then the equipment for routine maintenance can be planned to be procured late. Other areas like signage and graphics also need to be planned early. Operational

3.3 Financial Risk management measures

Planned project revenue should have, in addition to the fare box revenue, property development revenue, advertisement revenue and other forms.

There should be an all out effort to control the escalating prices. Cost control measures are needed to be applied. The major heads under which the costs can increase are – Alignment and Formation for underground section, Cost of land acquisition, change in cost of station buildings, Lifts and Escalators etc.

Commonly employed mechanisms for minimising completion risk before lending takes place include: (a) obtaining completion guarantees requiring the sponsors to pay all debts and liquidated damages if completion does not occur by the required date; (b) ensuring that sponsors have a significant financial interest in the success of the project so that they remain committed to it by insisting that sponsors inject equity into the project; (c) requiring parts of the project to be developed under fixed-price, fixed-time turnkey contracts by reputable and financially sound contractors whose performance is secured by performance bonds or guaranteed by third parties; and (d) obtaining independent experts' reports on the design and construction of the project. Completion risk is managed during the loan period by methods such as making pre-completion phase drawdowns of further funds conditional on certificates being issued by independent experts to confirm that the construction is progressing as planned

Metro rail projects are usually financed by limited recourse loan. A riskier or more expensive project may require limited recourse financing, secured by surety from sponsors. A complex project finance structure may incorporate corporate finance, securitization options, insurance provisions, or other types of collateral enhancement. While the borrower is in first loss position, the lender also assumes significant risk, so the lender must underwrite the loan with much more care than in a full recourse loan. This typically requires that the lender have significant domain expertise. For recourse financing, to ensure project revenues sufficient to service project debt, lenders can include in the provisions of the project debt documentation, a requirement that the sponsors inject the agreed funds into the project company to stabilize cash flow and reserves or top-up the shortfall in pre-completion project revenues

For future metro rail projects, a hybrid model of EPC and PPP modes is being recommended. With this, projects gain access to two critical aspects—the government's ability to address issues like land acquisition, raising funds with sovereign guarantee, and private sector entrepreneurship in bringing in technological and managerial experience.

Publicly-funded projects may also use additional financing methods such as tax increment financing or Private Finance Initiative (PFI). Such projects are often governed by a Capital Improvement Plan which adds certain auditing capabilities and restrictions to the process. New project finance structures may emerge primarily in response to the opportunity presented by long term power purchase contracts available from utilities and government entities.

3.4 Operational & Maintenance Risk management measures

Traffic projections need to be carried out for long horizons, about 10- 15 years from the first revenue operation date of the project. All provisions for space requirements in stations, the future requirements of rolling stock, enhancement of design loads and such areas need to be planned for.

If traffic demand and proper futuristic demand would be analyzed and projected accurately, the fare structure would be such that breakeven revenue can be generated early.

In developing countries, one or more government entities are the consumers of the project, and so purchase agreements between the government agencies and the project may contain clauses guaranteeing a minimum offtake and thereby guarantee a certain level of revenues. This serves to minimize the risks associated with traffic demand for the project investors and lenders.

If the Government is not the operator, the usual way for minimising operating risks before lending takes place is to require the project to be operated by a reputable and financially sound operator whose performance is secured by performance bonds.

Cash-flows are recommended to be controlled by requiring the proceeds of the sale of product to be paid into a tightly regulated proceeds account to ensure that funds are used for approved operating costs only.

Mechanisms for minimising resource risk should include: (a) matching the currencies of the sales contracts with the currencies of supply contracts as far as possible; (b) denominating the loan in

the most relevant foreign currency; and (c) requiring suitable foreign currency hedging contracts to be entered into.

Such resource risks are usually minimised by: (a) experts' reports as to the existence of the inputs (e.g. detailed reservoir and engineering reports which classify and quantify the reserves for a mining project) or estimates of public users of the project based on surveys and other empirical evidence (e.g. the number of passengers who will use a railway); (b) requiring long term supply contracts for inputs to be entered into as protection against shortages or price fluctuations (e.g. fuel supply agreements for a power station); (c) obtaining guarantees that there will be a minimum level of inputs (e.g. from a government that a certain number of vehicles will use a toll road); and (d) "take or pay" off-take contacts which require the purchaser to make minimum payments even if the product cannot be delivered

3.5 Concluding Remarks

It is felt that the measures recommended for planning of risk management in this paper can be useful for implementation on Indian metro railway projects, within the planning framework of that project. It is also recommended that a data base be created for quantifying risk related project data for the ongoing metro rail project. This will also enable quantitative risk analysis to be carried out.

Risk communication inside and outside the project organization ^{2 pp 389} will also be necessary. The use of RMIS (Risk Management Information System) ranging from simple spreadsheet based software to complex programs designed to organize, analyse and manage large amount of data on Risks can be helpful to project stakeholders.

5.0 References

1. Cooper Dale and Chapman Chris, 1987, , *Risk Analysis for large projects, models, methods and cases*, John Wiley
2. Young Peter C & Tippins Steven C. 2001, *Managing business risks*, AMACOM
3. RITES, May 2003, *Detailed Project Report– Bangalore Metro Phase I*
4. JICA, 2006, *Loan Agreement No. ID-P171 between JICA and Government of India*
5. 2006, MOUD, Metro rail Cell, *GOI approval letter for Bangalore Metro rail Project*

6. *Project Planning Documents* of Bangalore Metro Railway Project – project schedules, cost estimates, tender documents etc.
7. Project Management Institute, *PMBOK* 4th ed. 2008